Rohan Sagvekar

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EXPERIENCE

Data Analyst Intern Sep 2023 – Aug 2024

SRP BizExcell Pvt. Ltd. | Remote

- **Cut operational expenses by 30%** by running statistical analysis on business data to surface cost inefficiencies translated findings into structured data models and delivered recommendations directly to senior decision-makers.
- **Built 20+ Power BI dashboards** using SQL and Excel to automate end-to-end business intelligence reporting replaced manual reporting workflows and gave non-technical stakeholders real-time visibility into operational performance.
- Converted ambiguous business requirements into structured data models and clear visual reports, working cross-functionally with teams across the organization to align analytical outputs with operational priorities.

PROJECTS

ragprobe | Agentic AI Evaluation Framework (PyPI) Mar 2025 – Present

Python, OpenAI API, LangChain, Pydantic, SQLite, FAISS, GitHub Actions

- **Deployed an agentic AI workflow with tool-calling routines** that auto-generate adversarial test cases from any document corpus, reduced manual evaluation time by ~60%, and surfaced an 83% context-recall failure rate invisible to standard monitoring.
- Built a production monitoring layer with async scoring, SQLite trace logging, and threshold-based alerting, proactive detection of model failure modes across live financial corpora (AAPL, MSFT, NVDA) before they impacted outputs.
- Shipped full CI/CD via GitHub Actions with 260+ passing tests across Python 3.10–3.12; published to PyPI with a CLI supporting JSON, HTML, and Markdown output.

Gridwise | Fleet Telemetry Pipeline and Predictive Modeling Platform Jan 2024 – May 2024

Python, Flask, SQL, NumPy, pandas, Power BI, Matplotlib

- **Achieved 88% demand-prediction accuracy and 20% improvement in resource allocation efficiency** by developing time-series predictive models over live telemetry from 50+ distributed IoT sources surfacing degradation trends and usage patterns before they became field failures.
- **Built a scalable, automated data pipeline** for real-time fleet telemetry ingestion via a Flask REST API, with a SQL persistence layer for historical querying, enabling proactive monitoring across 50+ distributed field devices.
- Built visualizations Power BI dashboards for daily fleet monitoring and Matplotlib charts for statistical model performance to communicate forecast results to operational stakeholders; published at ICHCSC 2024.

Adversarial Robustness Research: Predictive Model Stress-Testing Jan 2025 – May 2025

PyTorch, Python, NumPy, Matplotlib

- **Quantified predictive model degradation from 96.3% to 0% accuracy** under adversarial distribution shift using numerical optimization across epsilon values, providing a reproducible benchmark for model reliability under real-world perturbations.
- **Recovered robust accuracy to 40.6% under PGD and 66.6% under FGSM,** a 40+ point gain by applying adversarial training (Madry et al., 2018) using linear algebra and probability-based attack formulations on a 60K-image dataset.
- Built visualizations, epsilon-vs-accuracy curves, and attack comparison charts to communicate the accuracy-vs-robustness trade-off to engineering reviewers without reading the code.

Real-Time Detection and Telemetry Extraction Pipeline Aug 2023 – Dec 2023

Python, YOLOv8, EasyOCR

- **Reduced manual review time by 40%** by engineering a real-time telemetry extraction pipeline processing 1,500+ feature extractions per session at 92% accuracy across varied field conditions.
- Deployed an automated alerting service with real-time anomaly flagging on matched detection events, replacing a human-in-the-loop monitoring workflow; published at ICNDA 2024 (Vol. 3).

TECHNICAL SKILLS

Languages: Python, SQL, Java, C/C++, R, JavaScript
Data Pipelines & Engineering: Flask, Fast API, REST APIs, SQLite, PostgreSQL, MySQL, Docker, Git, CI/CD (GitHub Actions), NumPy, pandas, Excel
AI Agents & LLMs: OpenAI API, LangChain, agentic workflows, tool-calling, RAG pipelines, LLM-as-Judge evaluation, prompt engineering, FAISS, Chroma
ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face Transformers, time-series forecasting, adversarial ML, linear algebra, numerical optimization
Visualization & Cloud: Power BI, Matplotlib, GCP, Azure, Docker, GitHub Actions, Linux

PUBLICATIONS

Sagvekar, R. et al. *"Gridwise: A Human-Centric Renewable Energy Monitoring Framework."* ICHCSC 2024.
Sagvekar, R. et al. *"Real-Time Vehicle Detection and License Plate Recognition Pipeline."* ICNDA 2024, Vol. 3.

EDUCATION

Stevens Institute of Technology Aug 2024 – May 2026
M.S., Data Science GPA: 3.67 / 4.0 | Hoboken, NJ
Ramrao Adik Institute of Technology Aug 2020 – May 2024
B. Tech, Computer Engineering (Minor: Data Science) GPA: 9.44 / 10 | Navi Mumbai, India